

Hospital AI Agent

Database Design

1. Introduction

The Hospital AI Agent system requires a database to store and manage information related to users, doctors, nurses, patients, appointments, departments, prescriptions, notifications, and hospital operations.

The database is designed to support the main hospital management functions as well as the AI Agent.

The AI Agent will retrieve authorized information from the database through controlled backend tools and APIs.

2. Main Database Entities

The main entities in the system are:

1. Users
 2. Departments
 3. Doctors
 4. Nurses
 5. Patients
 6. Appointments
 7. Prescriptions
 8. Medications
 9. Notifications
 10. Nurse Shifts
-

3. Users Table

The Users table stores general information for all system users.

Table Name: Users

Column	Data Type	Description
user_id	Integer	Primary Key
full_name	VARCHAR	User's full name
email	VARCHAR	User email
password_hash	VARCHAR	Encrypted password
phone	VARCHAR	Phone number
role	VARCHAR	User role
created_at	DATETIME	Account creation date

Possible Roles

- Admin
- Doctor
- Nurse
- Receptionist
- Patient

Primary Key

user_id

4. Departments Table

The Departments table stores hospital departments and medical specialties.

Table Name: Departments

Column	Data Type	Description
department_id	Integer	Primary Key
department_name	VARCHAR	Department name

Column	Data Type	Description
description	TEXT	Department description

Examples

- Cardiology
- Dermatology
- Pediatrics
- Orthopedics
- Neurology

Primary Key

department_id

5. Doctors Table

The Doctors table stores information related to doctors.

Table Name: Doctors

Column	Data Type	Description
doctor_id	Integer	Primary Key
user_id	Integer	Foreign Key
department_id	Integer	Foreign Key
specialization	VARCHAR	Doctor specialization
working_hours	VARCHAR	Doctor working hours
consultation_fee	DECIMAL	Appointment fee
status	VARCHAR	Doctor availability status

Primary Key

doctor_id

Foreign Keys

- user_id → Users.user_id
 - department_id → Departments.department_id
-

6. Patients Table

The Patients table stores patient information.

Table Name: Patients

Column	Data Type	Description
patient_id	Integer	Primary Key
user_id	Integer	Foreign Key
date_of_birth	DATE	Patient date of birth
gender	VARCHAR	Patient gender
emergency_contact	VARCHAR	Emergency contact
medical_history	TEXT	Medical history
created_at	DATETIME	Registration date

Primary Key

patient_id

Foreign Key

- user_id → Users.user_id
-

7. Nurses Table

The Nurses table stores information related to nurses.

Table Name: Nurses

Column	Data Type	Description
nurse_id	Integer	Primary Key

Column	Data Type	Description
user_id	Integer	Foreign Key
department_id	Integer	Foreign Key
assigned_doctor_id	Integer	Foreign Key
status	VARCHAR	Nurse status

Primary Key

nurse_id

Foreign Keys

- user_id → Users.user_id
- department_id → Departments.department_id
- assigned_doctor_id → Doctors.doctor_id

8. Appointments Table

The Appointments table is one of the most important tables in the system.

It connects patients with doctors.

Table Name: Appointments

Column	Data Type	Description
appointment_id	Integer	Primary Key
patient_id	Integer	Foreign Key
doctor_id	Integer	Foreign Key
appointment_date	DATE	Appointment date
appointment_time	TIME	Appointment time
status	VARCHAR	Appointment status
created_at	DATETIME	Creation date

Possible Status Values

- Booked
- Completed
- Cancelled

Primary Key

appointment_id

Foreign Keys

- patient_id → Patients.patient_id
 - doctor_id → Doctors.doctor_id
-

9. Prescriptions Table

The Prescriptions table stores prescriptions created by doctors.

Table Name: Prescriptions

Column	Data Type	Description
prescription_id	Integer	Primary Key
patient_id	Integer	Foreign Key
doctor_id	Integer	Foreign Key
appointment_id	Integer	Foreign Key
notes	TEXT	Doctor notes
created_at	DATETIME	Creation date

Primary Key

prescription_id

Foreign Keys

- patient_id → Patients.patient_id
- doctor_id → Doctors.doctor_id

- appointment_id → Appointments.appointment_id
-

10. Medications Table

The Medications table stores medication information associated with a prescription.

Table Name: Medications

Column	Data Type	Description
medication_id	Integer	Primary Key
prescription_id	Integer	Foreign Key
medication_name	VARCHAR	Medication name
dosage	VARCHAR	Required dosage
frequency	VARCHAR	Number of times
instructions	TEXT	Usage instructions
duration	VARCHAR	Treatment duration

Example

Medication Name: Medicine A

Dosage: 1 tablet

Frequency: Twice daily

Instructions: After meals

Duration: 7 days

Primary Key

medication_id

Foreign Key

- prescription_id → Prescriptions.prescription_id
-

11. Nurse Shifts Table

The Nurse Shifts table stores nurse working shifts.

Table Name: Nurse_Shifts

Column	Data Type	Description
shift_id	Integer	Primary Key
nurse_id	Integer	Foreign Key
department_id	Integer	Foreign Key
shift_date	DATE	Shift date
start_time	TIME	Shift start time
end_time	TIME	Shift end time
status	VARCHAR	Shift status

Possible Status Values

- Scheduled
- Active
- Completed

Primary Key

shift_id

Foreign Keys

- nurse_id → Nurses.nurse_id
- department_id → Departments.department_id

12. Notifications Table

The Notifications table stores notifications sent by the system.

Table Name: Notifications

Column	Data Type	Description
notification_id	Integer	Primary Key
user_id	Integer	Foreign Key
message	TEXT	Notification message
notification_type	VARCHAR	Type of notification
scheduled_time	DATETIME	Scheduled time
status	VARCHAR	Notification status
created_at	DATETIME	Creation date

Notification Types

- Appointment Reminder
- Shift Reminder
- Medication Reminder
- Administrative Alert

Possible Status

- Pending
- Sent
- Read

Primary Key

notification_id

Foreign Key

- user_id → Users.user_id

13. Main Relationships

The main relationships between the database tables are:

Users and Doctors

One User can be associated with one Doctor profile.

Users (1) → (1) Doctors

Users and Patients

One User can be associated with one Patient profile.

Users (1) → (1) Patients

Users and Nurses

One User can be associated with one Nurse profile.

Users (1) → (1) Nurses

Departments and Doctors

One Department can contain multiple Doctors.

Departments (1) → (Many) Doctors

Departments and Nurses

One Department can contain multiple Nurses.

Departments (1) → (Many) Nurses

Doctors and Appointments

One Doctor can have many Appointments.

Doctors (1) → (Many) Appointments

Patients and Appointments

One Patient can have many Appointments.

Patients (1) → (Many) Appointments

Doctors and Nurses

One Doctor may be assigned multiple Nurses.

Doctors (1) → (Many) Nurses

Appointments and Prescriptions

One Appointment may result in one or more Prescriptions.

Appointments (1) → (Many) Prescriptions

Prescriptions and Medications

One Prescription can contain multiple Medications.

Prescriptions (1) → (Many) Medications

Users and Notifications

One User can receive multiple Notifications.

Users (1) → (Many) Notifications

Nurses and Shifts

One Nurse can have multiple Shifts.

Nurses (1) → (Many) Nurse_Shifts

14. Simplified Database Structure

Users

↓

Doctors

Patients

Nurses

↓

Departments

↓

Appointments

↓

Prescriptions

↓

Medications

Additional Tables:

- Notifications
- Nurse_Shifts

15. Database Support for the AI Agent

The database will provide the information required by the AI Agent.

The AI Agent can use controlled backend tools to perform actions such as:

Search Doctor

The AI Agent searches the Doctors and Departments tables.

Check Available Appointment

The AI Agent checks the Appointments table and doctor schedules.

Book Appointment

The system creates a new record in the Appointments table.

Retrieve Patient Information

Authorized users can retrieve information from the Patients table.

View Medical History

Authorized healthcare professionals can access patient medical history.

Send Reminder

The system creates and manages reminders through the Notifications table.

Analyze Workload

The AI Agent can analyze:

- Number of appointments.
 - Doctor schedules.
 - Department workload.
 - Nurse assignments.
-

16. Database Security

The database must protect sensitive information.

The system should:

- Store passwords securely using password hashing.
 - Apply role-based access control.
 - Restrict access to patient medical information.
 - Allow the AI Agent to access data only through authorized tools.
 - Record important actions when required.
 - Prevent unauthorized modification of medical data.
-

17. MVP Database

For the first version of the project, the most important tables are:

1. Users
2. Departments
3. Doctors
4. Patients
5. Appointments

These tables are sufficient to build the first working version of the system.

The AI Agent can initially work with these tables to:

- Search for doctors.
- Find doctors by specialty.
- Check appointment availability.
- Suggest appointments.
- Assist with booking appointments.

Additional tables can be added gradually.

18. Conclusion

The database design provides the foundation for the Hospital AI Agent system.

It supports the main hospital management operations, including user management, doctor management, patient management, appointment scheduling, nurse management, prescriptions, medications, and notifications.

The database is designed to be scalable so that additional AI-powered features can be added in future development phases.